

E-Commerce — Foundations

Class Notes: Classes 01–03

Part I • Section EC-1 • Foundations of E-Commerce

CO4 — Apply Internet & E-Commerce Technologies

Bloom's: Understand → Analyze

Semester Final: 72 Marks

Theory: 39 Classes × 50 min

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1. Definition of E-Commerce

CORE DEFINITION

E-Commerce (Electronic Commerce) is the process of buying and selling goods, services, and information over electronic networks — primarily the Internet. It encompasses every digital transaction that involves the exchange of value between parties, whether B2B, B2C, or any other model.

E-Commerce is not merely an online shopping cart. It includes the entire ecosystem of digital commercial activity: product discovery, negotiation, payment, fulfillment, customer service, and after-sales interaction — all mediated by electronic systems.

SCOPE OF E-COMMERCE

E-Commerce covers:

- ▶ **Product transactions:** Physical goods ordered and paid for online (e.g., clothing on Daraz, books on Amazon)
- ▶ **Digital goods:** Software, music, ebooks, courses — delivered entirely digitally
- ▶ **Services:** Consulting, freelancing, financial services conducted via digital platforms
- ▶ **Information goods:** News subscriptions, databases, market reports sold online
- ▶ **Business processes:** Procurement, supply chain collaboration, banking — conducted electronically

EXAM TIP

A common exam question asks: "Define E-Commerce and distinguish it from E-Business." Memorize that E-Commerce = transactional (buying/selling), while E-Business = broader operational use of digital technology (covered in next point).

2. E-Commerce vs. E-Business

These two terms are often confused. The distinction is crucial:

Dimension	E-Commerce	E-Business
Scope	Transactional — buying and selling	Operational — all digital business processes

Dimension	E-Commerce	E-Business
Focus	Revenue-generating exchanges	Internal and external process optimization
Includes	Online storefronts, payment, digital contracts	E-Commerce + ERP, HR systems, internal communication, supply chain management
Example	Amazon allowing customers to buy a product online	Amazon's entire digital infrastructure: warehouse automation, logistics, HR systems, supplier portals
Relationship	Subset of E-Business	Superset — contains E-Commerce

KEY INSIGHT

All E-Commerce is E-Business, but not all E-Business is E-Commerce. A company using an internal ERP system for supply chain management is doing E-Business but not necessarily E-Commerce. A company accepting online payments is doing both.

3. Historical Evolution of E-Commerce

The development of e-commerce spans six decades and represents a fundamental reshaping of commerce:

○ 1960s — EDI Era

Electronic Data Interchange (EDI) emerged as the first form of electronic commerce. Large corporations (retailers, manufacturers) electronically exchanged standardized business documents — purchase orders, invoices, shipping notices — directly between computer systems, without human intervention. EDI required expensive private Value-Added Networks (VANs) and was only accessible to large enterprises. Companies like Walmart pioneered EDI to streamline supply chain communication with suppliers.

○ 1970s–1980s — Electronic Funds Transfer (EFT)

Banks began using EFT systems to move money between accounts electronically. SWIFT (Society for Worldwide Interbank Financial Telecommunication) was founded in 1973. Stock exchanges moved to electronic trading. Credit card networks expanded. These laid the financial infrastructure for future e-commerce.

○ 1991–1995 — Web 1.0 Era Begins

The World Wide Web was opened to commercial use in 1991. Tim Berners-Lee had invented HTML and HTTP in 1989–1990. In 1994, Netscape launched its browser and SSL (Secure Sockets Layer) for encrypted transactions. In 1995, Amazon launched as an online bookstore and eBay launched as a consumer-to-consumer auction platform. The first true commercial e-commerce sites appeared. Web 1.0 = static pages, read-only content, one-way information delivery.

○ 1995–2000 — Dot-com Boom

Massive investor speculation in internet companies. Thousands of e-commerce startups launched. Key launches: eBay (1995), Amazon (1995), PayPal (1998), Alibaba (1999), Google (1998). Many companies had astronomical valuations with no profits. The bubble burst in 2000–2001, wiping out hundreds of companies. Survivors (Amazon, eBay, Google) eventually became dominant global platforms.

○ 2004 — Web 2.0

Web 2.0 marked the shift from static, publisher-controlled content to user-generated, interactive, collaborative web. Key technologies: AJAX (asynchronous JavaScript), RSS feeds, APIs, wikis, social media. Facebook (2004), YouTube (2005), Twitter (2006) launched. E-commerce became social — user reviews, ratings, recommendations, and community-driven discovery. Amazon introduced user reviews; eBay relied on peer ratings. The web became a platform for participation, not just browsing.

○ 2007–2010 — Mobile Commerce (M-Commerce)

Apple launched the iPhone in 2007, fundamentally changing commerce. The App Store (2008) enabled native mobile shopping apps. Google launched Android. Mobile payments began with early NFC experiments. Customers could now browse, compare prices, and purchase from anywhere. Location-aware commerce (deals near you, store pickup) became possible. By 2010, smartphones were mainstream in developed markets.

○ 2010–Present — Social, Mobile, Cloud, Big Data

Instagram (2010) and Pinterest became shoppable platforms. Social commerce — selling directly through social media — exploded. China led with WeChat and Alibaba's ecosystem. Digital wallets (Apple Pay, Google Pay, bKash) replaced physical cards. AI-driven personalization (Amazon's recommendation engine, Netflix's algorithm) became standard. In Bangladesh: bKash (2011), Daraz (2012), Chaldal, and Shajgoj represent this era. TikTok Shop (2021+) introduced live-stream commerce.

EXAM TIP — FREQUENTLY TESTED

The evolution timeline is a very common short-answer or MCQ topic. Know: EDI (1960s) → Web 1.0 (1995) → Web 2.0 (2004) → Mobile/Social (2010+). Be able to name 1–2 key events per era.

4. Scale of Global E-Commerce

Understanding the sheer scale of e-commerce justifies its study as a major business discipline:

- ▶ **Global retail e-commerce sales** surpassed **USD 5.8 trillion** in 2023 (eMarketer), and are projected to reach \$8 trillion by 2027
- ▶ **China** is the world's largest e-commerce market, accounting for over 50% of global online retail sales. Alibaba's Singles' Day (11/11) alone generated over \$84 billion in 2021
- ▶ **United States:** Amazon generates ~\$500 billion annually, with AWS (cloud) subsidizing retail operations
- ▶ **Southeast Asia:** Fastest-growing e-commerce region; Shopee, Lazada, Tokopedia dominating

- ▶ **Bangladesh:** E-commerce market growing at ~20% annually; Daraz, Chaldal, bKash ecosystem accelerating growth
- ▶ **Mobile commerce:** Over 72% of all e-commerce transactions globally occur on mobile devices
- ▶ **Cross-border e-commerce:** 20–25% of global e-commerce now crosses national borders

5. Unique Features of E-Commerce (The Eight Features)

These eight features — extensively analyzed by Laudon & Traver — distinguish e-commerce from traditional commerce. Each is a potential exam question:

Feature	Definition	Example	Business Implication
Ubiquity	E-commerce is available everywhere and at all times — not limited to a physical marketplace	Shopping at 2 AM from your bedroom via Daraz app	Extends market space; consumers can shop anytime, reducing time cost of purchase
Global Reach	Commerce can cross national and cultural boundaries with minimal incremental cost	A Bangladeshi artisan selling handloom products to a German buyer via Etsy	Massively expands the potential market; SMEs can reach global customers without offices abroad
Universal Standards	Internet uses globally shared technical standards (TCP/IP, HTTP, HTML) so any device can interact with any website	A website built in Bangladesh works identically for a user in Japan	Drastically reduces market entry costs; no proprietary technology lock-in like pre-web EDI
Richness	Information delivered can be rich: video, audio, animation, images, text — far beyond what a printed catalogue or TV ad can offer	360-degree product views, instructional videos on a product page, live chat	Reduces information asymmetry; merchants can communicate product value as effectively as a physical salesperson
Interactivity	E-commerce technology allows two-way communication between merchant and consumer; it responds to user input	Chatbots answering questions, product configurators, personalized homepages	Simulates face-to-face selling at global scale; enables mass customization

Feature	Definition	Example	Business Implication
Information Density	The internet gives buyers and sellers access to a huge amount of information very easily and cheaply	Price comparison websites (Shopper.com.bd), product review aggregators	Reduces information asymmetry; empowers consumers; forces price transparency on merchants
Personalization / Customization	Modern technology can show different products, ads, or messages to different people depending on their personal data.	Amazon's "Customers who bought this also bought..." section; Netflix recommendations	Increases conversion rates; creates switching costs; enables relationship marketing at scale
Social Technology	Internet enables users to create and share content in large social networks; commerce becomes embedded in social environments	Instagram Shops, Facebook Marketplace, TikTok Shop, WhatsApp Business	User-generated content (reviews, unboxing videos) drives purchasing decisions; community creates trust

6. E-Commerce vs. Traditional Commerce

Dimension	Traditional Commerce	E-Commerce
Time	Limited to business hours; seasonal sales events	24/7/365 availability; no downtime
Geography	Limited to physical location; local/regional market	Global reach; any buyer, any seller, any location
Cost	High fixed costs: rent, staff, physical inventory display	Lower fixed costs; digital catalogue is nearly free to maintain; variable cost model
Reach	Footfall-dependent; customer must come to you	You reach customers through search, social media, ads
Inventory	Physical stock must be on display	Dropshipping and virtual inventory possible; sell before you stock
Price transparency	Price negotiation; price hidden from competitor view	Real-time price comparison; price wars common
Customer data	Minimal data; loyalty cards at best	Rich behavioral data: every click, search, dwell time is recorded
Marketing	Mass media: TV, print, radio	Targeted digital marketing: SEO, PPC, email, social

Dimension	Traditional Commerce	E-Commerce
Transaction speed	Manual processing; delays in payment and confirmation	Real-time order confirmation, digital payment, instant delivery for digital goods

CASE CONTEXT: BANGLADESH

In Bangladesh, traditional commerce through haats (village markets), permanent shops, and bazaars still dominates rural areas. However, bKash mobile payments and platforms like Daraz and Chaldal have brought e-commerce to urban and semi-urban populations. The shift is accelerating due to smartphone penetration and 4G coverage expansion under Digital Bangladesh policy.

WHAT IS A BUSINESS MODEL?

A **business model** is a plan for how an organization creates, delivers, and captures value. In e-commerce, the business model describes who the participants are (buyer/seller), how value flows between them, and how money is made. Understanding business models is foundational to analyzing any e-commerce company.

1. B2C — Business-to-Consumer

B2C is the most familiar form of e-commerce: a business sells products or services directly to individual consumers through a digital channel.

B2C SUB-CATEGORIES

- ▶ **Online Retailers (e-tailers):** Digital storefronts selling physical or digital goods. Inventory-based model — the company owns or warehouses the goods. Example: *Daraz.com.bd* (Bangladesh's dominant marketplace), *Amazon* (US/global), *Flipkart* (India). Revenue = product margin + marketplace commissions.
- ▶ **Portals:** Web portals (Yahoo, AOL historically) provide aggregated content and services — news, email, search, entertainment — and monetize via advertising. The "portal" attracts traffic; advertisers pay for exposure.
- ▶ **Content Providers:** Sell access to digital content — news, streaming, academic journals. Examples: *Netflix* (subscription streaming), *Prothom Alo* digital (news), *Coursera* (courses). Revenue: subscription or per-item purchase.
- ▶ **Transaction Brokers:** Facilitate transactions between buyers and sellers, earning a commission without owning the product. Examples: *Booking.com* (hotel bookings), *Shohoz* (Bangladesh ticketing). Revenue: transaction fee or commission.
- ▶ **Market Creators:** Build digital marketplaces where third-party buyers and sellers transact. The platform itself owns no inventory. Examples: *Airbnb* (accommodation), *Uber* (rides), *eBay*. Revenue: commission on each transaction.
- ▶ **Service Providers:** Deliver services online — legal advice, financial planning, telemedicine, cloud storage. Examples: *Google Drive*, *Dropbox*, Bangladesh's *Doctorola* for telemedicine.

Daraz operates as both an online retailer (Daraz's own inventory) and a market creator (third-party sellers list on Daraz's platform). This hybrid model is common among major e-commerce platforms globally.

2. B2B — Business-to-Business

B2B e-commerce involves commercial transactions between businesses. It is actually the *largest* form of e-commerce by volume globally — far exceeding B2C — because every supply chain interaction between manufacturers, distributors, and wholesalers is potentially a B2B e-commerce transaction.

B2B SUB-CATEGORIES

- **Net Marketplaces (e-marketplaces):** Digital platforms where many buyers and many sellers in an industry transact. Examples: *Alibaba.com* (global sourcing marketplace — manufacturers meet international buyers), *IndiaMART*. Alibaba connects Chinese factories with global importers. Revenue model: membership fees + advertising + transaction commissions.
- **Private Industrial Networks:** Single large company sets up a private extranet connecting it to its specific suppliers and partners. The network is invitation-only, not open to all. Example: *Walmart's Retail Link* — all Walmart suppliers must connect to this proprietary platform to receive orders and submit invoices. Highly efficient but creates dependence on the dominant buyer.
- **E-Procurement Systems:** Companies use software to automate their purchasing processes — purchase requisitions, vendor selection, order placement, invoice matching. Examples: *SAP Ariba*, *Coupa*. Reduces procurement cycle time and cost.
- **Exchanges:** Real-time B2B markets for commodity goods (steel, chemicals, energy). Prices fluctuate based on supply and demand. Less common after the dot-com era as many failed, but some industry-specific exchanges survive.

Feature	B2C	B2B
Transaction volume	High number of small transactions	Fewer but much larger transactions
Decision making	Individual, emotional, impulsive possible	Committee-based, rational, ROI-driven
Relationship	Relatively anonymous, transactional	Long-term, contract-based relationships
Price	Fixed listed price	Negotiated pricing, volume discounts
Catalog complexity	Consumer-friendly, visual	Technical specifications, SKU codes
Payment terms	Immediate payment	Net 30/60/90 day payment terms

3. C2C — Consumer-to-Consumer

In C2C, individual consumers sell to other individual consumers, typically enabled by a platform that provides infrastructure, trust mechanisms (ratings, dispute resolution), and payment processing.

- ▶ **Auction platforms:** *eBay* — the original C2C auction site. Sellers list items; buyers bid. Highest bidder wins. Revenue: listing fees + final value fee (% of sale price). Now also includes fixed-price listings.
- ▶ **Classified platforms:** *OLX* (operates in Bangladesh as well) — free listings for local buyers and sellers. Like a digital newspaper classifieds section. Revenue: premium listing fees, featured ads.
- ▶ **Social commerce:** *Facebook Marketplace* — massive peer-to-peer local marketplace built into Facebook. Facebook does not charge transaction fees (yet) but leverages the activity for ad targeting data.
- ▶ **Peer review platforms:** Though not strictly transactional, sites like *TripAdvisor* and *Yelp* create consumer-generated content that enables C2C information sharing that drives commerce.

TRUST PROBLEM IN C2C

The fundamental challenge of C2C is trust. Unlike buying from a reputable brand (B2C), you are buying from a stranger. Platforms address this through rating and review systems, verified seller badges, buyer protection programs, and escrow payment. *eBay's* 5-star rating system is the canonical solution.

Facebook Marketplace relies on social identity as a trust signal.

4. C2B — Consumer-to-Business

C2B is a relatively modern model where *individuals* sell products or services to businesses. The power relationship is reversed: instead of a business marketing to consumers, the individual offers value to a business.

- ▶ **Freelancing platforms:** *Upwork*, *Fiverr* — individuals (consumers) offer skills (web development, design, writing, accounting) to businesses. Bangladesh is a top-5 global freelancing country. Freelancers earn foreign currency, contributing significantly to remittances.
- ▶ **Reverse auctions:** Consumers name their price for a product/service and businesses decide whether to accept. *Priceline* pioneered this for hotel rooms and airline tickets.
- ▶ **Influencer marketing:** Individuals (influencers) charge businesses for sponsored posts, product reviews, or brand mentions on their social media channels. An individual's audience becomes a product sold to brands.
- ▶ **Data/Feedback provision:** Individuals participate in paid surveys, product testing, UX research — selling their opinions and time to businesses.
- ▶ **User-generated content licensing:** Stock photo websites (*Shutterstock*, *Getty Images*) buy photographs from individual photographers and resell them to businesses.

5. B2G and G2C — Government Models

B2G — BUSINESS-TO-GOVERNMENT

Businesses sell goods and services to government agencies through digital channels. Key example: **e-tendering / e-procurement** systems where government departments publish tenders online and businesses submit bids digitally. Bangladesh's Central Procurement Technical Unit (CPTU) operates the e-Government Procurement (e-GP) portal — all government tenders above a threshold must be published here. This reduces corruption and increases transparency.

G2C — GOVERNMENT-TO-CITIZEN

Government provides services to citizens digitally:

- ▶ **Digital public services:** Online passport application, NID services, land record lookup, birth certificate requests
- ▶ **e-Tax filing:** Online income tax submission (NBR's e-Filing portal in Bangladesh)
- ▶ **Utility payments:** Online electricity bill payment (DESCO, NESCO), water bill
- ▶ **Digital Bangladesh initiative:** 5,000+ Union Digital Centers (UDCs) bringing government digital services to rural populations

6. M-Commerce — Mobile Commerce

M-Commerce refers to commercial transactions conducted through mobile devices (smartphones, tablets). It is not a separate business model but rather a *channel* through which any of the above models (B2C, B2B, C2C, etc.) can operate.

- ▶ **Mobile shopping apps:** Dedicated native apps for retailers (Daraz app, Amazon app). Apps typically have higher conversion rates than mobile websites because of push notifications, stored payment info, and smoother UX.
- ▶ **Mobile payments:** *bKash* (Bangladesh), *Nagad* — Mobile Financial Services (MFS). Users transfer money, pay merchants, pay utility bills, receive remittances via mobile app or USSD code. bKash has 60+ million registered users in Bangladesh.
- ▶ **Super-apps:** WeChat (China), Grab (Southeast Asia) — single apps that combine messaging, payments, food delivery, ride-hailing, e-commerce, and financial services. Bangladesh's Pathao and Shohoz are moving toward super-app models.
- ▶ **Location-based commerce:** Apps use GPS to show nearby deals, stores, or services. Google Maps "restaurant nearby" + order integration is an example.

7. Social Commerce

Social Commerce means people can shop directly through social media apps like Facebook, Instagram, TikTok, etc

- ▶ **Shoppable posts:** Instagram and Pinterest allow businesses to tag products in posts. Users see a product in a lifestyle photo and can purchase without leaving the app.
- ▶ **Live-stream commerce:** Sellers demonstrate products in real-time video streams while viewers purchase during the broadcast. Dominant in China (Taobao Live, Douyin). Growing rapidly in Bangladesh via Facebook Live.
- ▶ **Influencer-driven sales:** Influencers share affiliate links. When followers purchase through the link, the influencer earns a commission. TikTok Shop formalizes this with in-app purchasing.
- ▶ **Community group buying:** Groups of buyers aggregate demand for volume discounts. *Pinduoduo* (China) built a \$100B+ company on this model. Bangladesh has informal WhatsApp group buying.
- ▶ **Social proof mechanisms:** Number of purchases ("500+ bought today"), star ratings, user photos — social signals embedded in product pages to influence purchase decisions.

8. Comprehensive Model Comparison Table

Model	Participants	Revenue Mechanism	Primary Examples	Bangladesh Examples
B2C	Business → Consumer	Product margin, subscriptions, ads	Amazon, Netflix, Walmart.com	Daraz, Chaldal, Shajgoj
B2B	Business → Business	Contracts, volume pricing, SaaS fees	Alibaba, SAP Ariba, Grainger	e-GP portal, local wholesalers
C2C	Consumer → Consumer	Platform listing/transaction fees	eBay, OLX, Facebook Marketplace	OLX BD, Facebook Marketplace BD
C2B	Consumer → Business	Project fees, commissions, data fees	Upwork, Fiverr, Shutterstock	Upwork (major BD export earner)
B2G	Business → Government	Government contracts, tenders	SAP (govt ERP), Cisco (networking)	e-GP portal, ICT procurement
G2C	Government → Citizen	Service fees (minimal), efficiency gains	IRS e-file, HMRC digital	NID services, e-Tax, UDCs
M-Commerce	Any model via mobile	Same as base model + mobile-specific	Amazon app, WeChat Pay	bKash, Nagad, Daraz app
Social Commerce	Any model via social	Commission, ads, affiliate fees	Instagram Shop, TikTok Shop	Facebook Live sellers

EXAM TIP — HIGH PROBABILITY QUESTION

"Classify the following scenarios into the correct e-commerce business model" is a standard exam question. For each model, know: (1) who the buyer and seller are, (2) at least one global example, (3) one Bangladesh-specific example. The C2B model (reverse flow) and social commerce are frequently tested because students confuse them with B2C.

1. E-Commerce Revenue Models

DEFINITION

A **revenue model** describes how a firm generates income. It is a key component of the overall business model. E-commerce firms often combine multiple revenue models simultaneously — this is called a **multi-sided revenue model**.

Advertising Model

Revenue from advertisers

The firm provides free content/services to attract a large user base, then charges advertisers to reach that audience. The user is the product; attention is what is sold.

Examples: Google Search ads (CPC), Facebook/Instagram ads, YouTube pre-roll, Prothom Alo digital ads. Google earns ~77% of revenue from advertising.

Subscription Model

Recurring periodic payments

Users pay a fixed fee at regular intervals (monthly, annually) for access to a service or content. Creates predictable, recurring revenue. High customer lifetime value if churn is low.

Examples: Netflix (\$15–\$22/month), Spotify, LinkedIn Premium, Adobe Creative Cloud (\$55/month). SaaS software universally uses this model.

Transaction Fee Model

Commission per transaction

The platform charges a percentage of each transaction it facilitates. No transaction = no revenue. Aligns platform incentives with user success.

Examples: eBay (10–15% final value fee), Upwork (20% on first \$500 per client), Airbnb (3% host fee + 14% guest fee), stock brokerages. Daraz charges sellers 3–10% commission.

Freemium Model

Free base + paid premium

Core product is free to attract large user base; advanced features require payment. Conversion from free to paid is typically 2–5%. The free tier also serves as marketing.

Examples: Spotify (free with ads vs. premium ad-free), Dropbox (2GB free / paid for more storage), LinkedIn (free profile / Sales Navigator), Zoom (40-min limit free / paid unlimited).

Affiliate / Referral Model

Commission for referred traffic/sales

Affiliate publishers (bloggers, websites, influencers) include tracked links to a merchant. When a user

Sales / E-tail Model

Direct product margin

The classic retail model: buy at wholesale price, sell at retail price, keep the margin. Straightforward but

clicks and purchases, the affiliate earns a commission (typically 4–10% of sale value).

Examples: Amazon Associates (world's largest affiliate program — millions of publishers earn by linking to Amazon products), booking.com affiliates, Daraz affiliate program.

requires inventory management, capital, and logistics. Margin typically 20–60%.

Examples: Chaldal (online grocery), any direct-to-consumer brand (DTC). Apple earns massive margins selling hardware through its own website.

AMAZON'S MULTI-REVENUE-MODEL ECOSYSTEM (CASE STUDY)

Amazon is the canonical example of combining multiple revenue models into a dominant platform:

- ▶ **E-tail / Sales:** Amazon.com sells 1st party products (Amazon owns inventory) — product margin revenue
- ▶ **Marketplace (Transaction Fee):** 60%+ of Amazon's unit sales come from 3rd party sellers; Amazon charges 8–15% referral fee per sale
- ▶ **Subscription:** Amazon Prime (~200 million members globally at \$139/year) provides free shipping, video, music — and increases purchase frequency dramatically
- ▶ **Advertising:** Amazon Advertising is the 3rd largest digital ad platform globally — sellers pay to appear prominently in search results
- ▶ **AWS (IaaS):** Amazon Web Services — cloud infrastructure sold to businesses. AWS accounts for ~75% of Amazon's total operating profit despite being ~15% of revenue
- ▶ **Affiliate:** Amazon Associates program — millions of websites link to Amazon products

ANALYTICAL INSIGHT — CROSS-SUBSIDIZATION

Amazon's highly profitable AWS business effectively cross-subsidizes its razor-thin retail margins. This allows Amazon to offer competitive prices that brick-and-mortar retailers cannot match. Understanding cross-subsidization between revenue streams is essential to analyzing platform business models.

2. Value Chain Analysis in E-Commerce

DEFINITION — PORTER'S VALUE CHAIN

Michael Porter's **Value Chain** model (1985) analyzes the activities a firm performs to create and deliver a product/service to customers. Each activity either adds value or adds cost. Competitive advantage comes from performing activities better or more cheaply than competitors.

PRIMARY ACTIVITIES (DIRECTLY CREATE VALUE)

1. Inbound Logistics Receiving, storing, distributing inputs. E-commerce: supplier portals, automated inventory receiving, barcode/RFID scanning of incoming stock.	2. Operations Transforming inputs into outputs. E-commerce: order processing systems, warehouse management software, automated picking robots (Amazon Kiva).	3. Outbound Logistics Delivering products to customers. E-commerce: last-mile delivery, same-day/next-day fulfillment, returns management, tracking systems.	4. Marketing & Sales Creating awareness and driving purchases. E-commerce: SEO, SEM, email marketing, social media, recommendation engines, personalized ads.	5. Service Post-sale support. E-commerce: chatbots, online returns portals, knowledge bases, customer reviews/Q&A systems, loyalty programs.
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SUPPORT ACTIVITIES (ENABLE PRIMARY ACTIVITIES)

- Firm Infrastructure:** Finance, legal, management information systems, accounting. E-commerce: ERP systems, business intelligence dashboards, cloud infrastructure.
- Human Resource Management:** Recruiting, training, compensation. E-commerce requires data scientists, software engineers, UX designers, digital marketers — specialized talent.
- Technology Development:** R&D, software development, process innovation. For e-commerce firms, this is a primary differentiator — Amazon spends billions annually on technology.
- Procurement:** Purchasing inputs (products, raw materials, services). E-commerce: automated e-procurement, vendor management portals, reverse auctions for supplier selection.

INDUSTRY VALUE CHAIN VS. FIRM VALUE CHAIN

The **industry value chain** (also called the "supply chain" in common usage) describes the sequence of firms involved in creating and delivering a product from raw material to end consumer:

Raw Materials Farmers, miners, basic material producers	Manufacturers Convert materials into finished goods	Wholesalers Aggregate products from multiple manufacturers	Distributors Regional distribution to retailers	Retailers Sell to consumers (physical or digital)
Consumers End users				

The **firm's value chain** is the set of activities within a single company in that chain. E-commerce disrupts the industry value chain by enabling firms to skip intermediaries (disintermediation).

3. Disintermediation and Re-intermediation

Disintermediation is the removal of intermediaries (middlemen) from a supply chain as a result of digital connectivity. When a manufacturer can sell directly to consumers online, wholesalers and distributors lose their role.

Examples of disintermediation in e-commerce:

- ▶ **Music industry:** Spotify/Apple Music eliminated music retailers (HMV, Tower Records). Artists can now distribute on Spotify directly, bypassing physical distributors.
- ▶ **Travel industry:** Expedia, Booking.com, and airline websites eliminated most travel agencies. Consumers book directly. 80%+ of airline tickets are now sold online.
- ▶ **Book publishing:** Amazon KDP (Kindle Direct Publishing) allows authors to self-publish and sell ebooks directly, bypassing traditional publishers and bookstores.
- ▶ **Bangladesh garments:** Some manufacturers use B2B portals to connect with international buyers directly, reducing reliance on buying agents.

RE-INTERMEDIATION

Re-intermediation is the creation of new types of intermediaries that perform value-adding functions in the digital economy — even as old intermediaries are eliminated. The internet removes some middlemen but creates entirely new ones.

- ▶ **Price comparison engines:** Google Shopping, PriceRunner — new intermediaries that aggregate prices from multiple retailers, helping consumers find the best deal
- ▶ **Marketplace platforms:** Amazon, Alibaba — these are new intermediaries between buyers and third-party sellers
- ▶ **Payment processors:** PayPal, Stripe, bKash — new financial intermediaries for digital transactions
- ▶ **Review aggregators:** Trustpilot, Google Reviews — intermediaries that aggregate consumer trust signals
- ▶ **Logistics aggregators:** Pathao Courier, Paperfly (Bangladesh) — intermediaries connecting merchants with multiple delivery services

KEY INSIGHT

Disintermediation and re-intermediation often happen simultaneously in the same industry. Travel is a perfect example: travel agencies were eliminated (disintermediation) but Booking.com, Expedia, and Google Travel emerged as powerful new digital intermediaries (re-intermediation) who charge the same or higher fees than the agents they replaced.

4. Network Effects and Metcalfe's Law

NETWORK EFFECT (DEFINITION)

A **network effect** (also called network externality) occurs when a product or service becomes more valuable as more people use it. The value of the network to each user increases with every new user who joins.

METCALFE'S LAW

$$V \propto n^2$$

where V = value of the network, n = number of users

Metcalfe's Law states that the value of a telecommunications or social network is proportional to the square of the number of connected users. Intuition: with 2 users, there is 1 possible connection. With 5 users, there are 10 connections. With 100 users, there are 4,950 possible connections. Value grows exponentially while the cost of adding users grows linearly.

TYPES OF NETWORK EFFECTS

- ▶ **Direct network effects:** Value comes directly from other users of the same product. Examples: telephone network (useless alone, valuable when everyone has it), WhatsApp (more valuable when your contacts use it), Facebook (more valuable when your friends are on it)
- ▶ **Indirect (two-sided) network effects:** Value comes from related but different user groups. More sellers on eBay attract more buyers; more buyers attract more sellers. More app developers on iOS attract more iPhone buyers; more iPhone buyers attract more developers. This is the foundation of platform economics.
- ▶ **Data network effects:** The more users a platform has, the more data it collects, which improves its AI/recommendation engine, which attracts more users. Amazon's recommendation engine improves with every purchase; Google's search improves with every query. This creates a data moat.

EXAM TIP — METCALFE'S LAW CALCULATION

You may be asked: "A network has 10 users. If users double to 20, by what factor does network value increase?" Answer: Value is proportional to n^2 . So: $10^2 = 100$ (original) and $20^2 = 400$ (new). Value increases by a factor of 4 (4×). Users doubled but value quadrupled. This demonstrates the power law growth of network effects.

5. Platform Economics and Two-Sided Markets

TWO-SIDED MARKET

A **two-sided market** (also: multi-sided platform) connects two or more distinct user groups who benefit from interacting with each other. The platform provides the infrastructure for this interaction and monetizes it. The key insight is that one side's participation makes the platform more attractive to the other side.

CLASSIC TWO-SIDED PLATFORM EXAMPLES

Platform	Side 1	Side 2	How Value Flows	Revenue Model
Daraz	Buyers/consumers	Sellers/merchants	More buyers → more sellers → more product variety → more buyers	Commission from sellers; ads from sellers to buyers
Google Search	Search users (free)	Advertisers (paid)	More users → more valuable ads → more revenue → better product → more users	Cost-per-click advertising (100% from advertisers)
Airbnb	Travelers/guests	Hosts/property owners	More listings → more travel bookings → more hosts join	% fee from both sides per booking
Uber/Pathao	Riders	Drivers	More drivers → shorter wait times → more riders → more earnings for drivers	Commission from driver per trip
Apple App Store	iPhone users	App developers	More users → more developer investment → better apps → more users	30% commission on app sales/subscriptions

DEMAND AGGREGATION

Demand aggregation is the ability of digital platforms to pool together small, dispersed demands from many consumers into collectively significant volumes. Before the internet, a niche product (e.g., left-handed guitars) might not have enough local demand to support a retailer in any given city. Amazon aggregates global demand for niche products, enabling the "long tail" — vast selection of niche products, each selling in low volume but collectively generating enormous revenue.

Chris Anderson's Long Tail theory (2004): In the digital economy, the "long tail" (thousands of niche products, each with low individual demand) collectively outperforms the "short head" (a few blockbuster products). Netflix, Spotify, and Amazon all profit from the long tail of content and products.

6. Winner-Takes-All Dynamics

Network effects create a powerful tendency toward market concentration in digital markets — the biggest platform attracts the most users, which makes it even more attractive to new users, which gives it even more data to improve its service. This creates a **winner-takes-all** (or winner-takes-most) dynamic:

- ▶ **Search:** Google has ~93% global search market share. Bing/Yahoo together have ~6%. New entrants face an enormous data disadvantage.
- ▶ **Social:** Facebook/Instagram/WhatsApp dominate because your friends are there. Switching costs are social, not financial.
- ▶ **Ride-sharing:** Uber/Lyft in US; in Bangladesh, Pathao and Shohoz compete, but consolidation is expected.
- ▶ **E-commerce:** Amazon has ~40% of US e-commerce. Daraz has dominant position in Bangladesh.

LOCK-IN EFFECTS

Platforms create **lock-in** — high switching costs that make it difficult for users to leave even if a competitor offers a marginally better product. Types: (1) *Data lock-in* — your purchase history, playlists, settings are only on one platform; (2) *Network lock-in* — your friends/followers are on one platform; (3) *Ecosystem lock-in* — Apple hardware + iOS + iCloud + App Store purchases all reinforce each other.

7. Summary: Key Concepts for Exam Review

Concept	Core Definition (in your own words for exam)	Key Formula or Example
Revenue Model	How a firm earns money	Advertising, subscription, transaction fee, freemium, affiliate, sales
Value Chain	Sequence of activities a firm performs to create and deliver value	Primary: inbound logistics → operations → outbound logistics → marketing → service; Support: infrastructure, HRM, tech, procurement
Industry Value Chain	All firms in the supply chain from raw material to consumer	Supplier → Manufacturer → Wholesaler → Retailer → Consumer
Disintermediation	Removal of middlemen due to digital connectivity	Travel agencies eliminated by Booking.com / airline websites
Re-intermediation	Creation of new digital middlemen to replace old ones	Booking.com replaces travel agents but is itself an intermediary
Network Effect	Product becomes more valuable as more users join	WhatsApp, Facebook, Uber driver/rider network

Concept	Core Definition (in your own words for exam)	Key Formula or Example
Metcalfe's Law	Network value grows as the square of number of users	$V \propto n^2$ — double users = quadruple value
Two-Sided Market	Platform connecting two distinct user groups who need each other	Daraz: buyers + sellers; Google: users + advertisers
Demand Aggregation	Pooling dispersed small demands into significant volumes	Amazon long tail — niche products viable at global scale
Winner-Takes-All	Network effects drive market concentration toward 1–2 dominant players	Google 93% search share; Amazon 40% US e-commerce
Lock-in	High switching costs trap users in a platform's ecosystem	Apple ecosystem: hardware + software + content + services

FINAL EXAM STRATEGY — CLASSES 01 TO 03

For essay questions: Structure answers using (1) definition with precision, (2) key features or components, (3) real-world examples (both global and Bangladesh-specific), (4) analytical comparison or implication. For MCQs: focus on exact definitions, the 8 unique features of e-commerce, business model classifications with concrete examples, and Metcalfe's Law formula. The Amazon multi-revenue-model case is likely to appear as a structured case question. Practice distinguishing disintermediation from re-intermediation with examples.